

CHINA MATTERS

All About Tech

- China's May activity data continued to disappoint outside of exports and industrial production. With export growth contributing roughly 3pp to year-over-year real GDP growth, domestic demand appears to be growing at a sluggish pace of only 1-2% yoy in recent months based on official data.
- We nudge down our Q2 real GDP forecast from 4.0% qoq annualized to 3.5% on the back of weak April and May data, implying year-over-year growth of 4.5% (vs. 4.7% previously). Assuming lower oil prices, faster fiscal spending, and normalized weather conditions, we project Q3 real GDP growth to rebound to 5.0% qoq annualized (vs. 4.5% previously). Our full-year real GDP growth forecast remains unchanged at 4.7%.
- The Chinese government focuses on its long-term objective of transforming the economy into a tech-driven one. The divergence between high-tech/AI and property/consumption continues to widen in both industrial production and capital market data. Top leaders' domestic travel, recent policy communications, and our on-the-ground channel checks all suggest these trends will persist.
- While China's long-term planning has proved valuable in weathering external shocks such as the US-China trade war and energy price spikes, we believe cyclical policies are crucial to ensuring a smoother path toward China's strategic goals. We discuss three channels connecting near-term policies with long-term objectives.
- First, AI-related job displacement could amplify macroeconomic headwinds and delay, if not derail, the recovery in the property market and household consumption. Second, with export growth likely to remain strong over the next few years, targeting only headline GDP growth could cap domestic demand growth. Third, frequent quarter-to-quarter shifts in the policy stance to steer the economy toward the full-year target are not conducive to confidence-building. If not managed well, persistent cyclical weakness could impede China's goal of restructuring the economy into a virtuous cycle between tech-driven productivity gains and strong domestic demand growth.

Hui Shan

+852-2978-6634 | hui.shan@gs.com
Goldman Sachs (Asia) L.L.C.

All About Tech

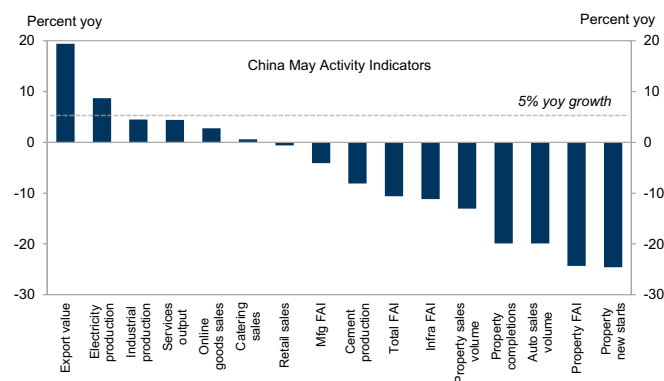
An even more divergent economy

China's May activity data disappointed again following weak April readings. Both retail sales and fixed asset investment (FAI) contracted: May retail sales declined 0.6% yoy, and January-May FAI dropped 4.1% yoy. In the history of official releases, this is only the second time on record that both retail sales and FAI have printed negative numbers—the first was in 2020 during Covid lockdowns.

Looking across major indicators, exports remain the best-performing sector, rising nearly 20% yoy in nominal USD terms. By contrast, domestic indicators such as property activity, auto sales, and infrastructure investment (on a single-month basis) showed double-digit year-over-year declines (Exhibit 1). Auto sales data best illustrate the significant divergence between exports and domestic demand: domestic sales dropped 22% yoy in May, while auto exports jumped 75% yoy (Exhibit 2).

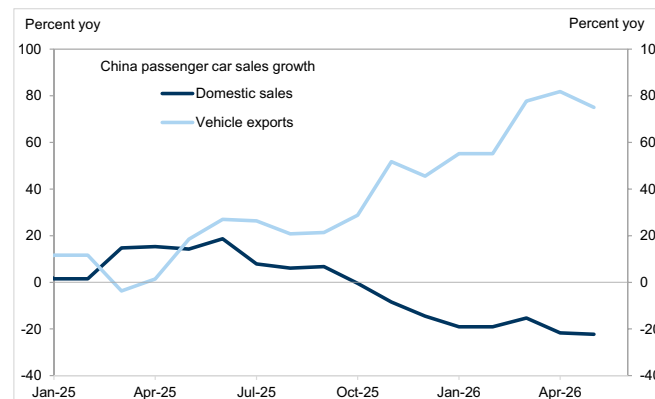
The April and May production data track roughly 4% yoy real GDP growth, about 1pp lower than the Q1 official real GDP growth of 5%. We calculate that the ongoing robust export growth likely contributes around 3pp to year-over-year real GDP growth, implying that domestic demand is growing at a very sluggish pace of only 1-2% yoy based on official statistics. This is also consistent with our import-implied domestic demand growth estimates.

Exhibit 1: Exports remain the best performing part of the economy



Source: CEIC, Goldman Sachs Global Investment Research

Exhibit 2: Domestic sales of autos dropped 22% yoy in May while exports jumped 75% yoy



Source: Haver Analytics, Goldman Sachs Global Investment research

A Slower Q2 and a Faster Q3

Given the April and May realized data, we nudge down our Q2 real GDP growth forecast from 4.0% to 3.5% in qoq annualized terms, or from 4.7% to 4.5% in year-over-year terms. The sequential slowdown from Q1 to April-May was likely driven by three factors.

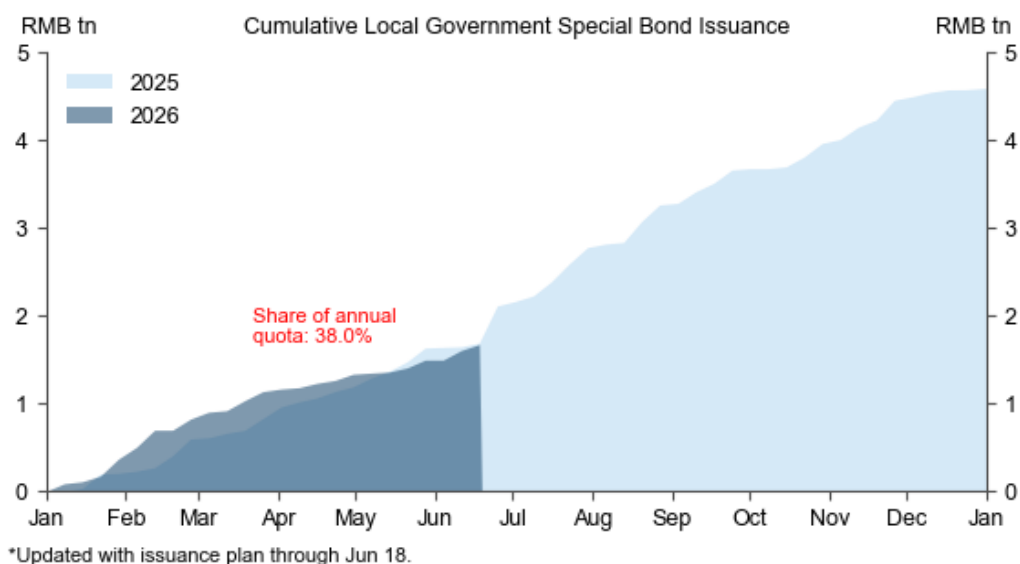
- 1. The negative impact of higher energy prices:** Notable declines in the production of refined oil and chemicals, as well as the sharp drop in domestic retail fuel sales, are clear signs that the Iran war and higher oil prices weighed on Chinese economic activity. With the interim deal to reopen the Strait of Hormuz and our commodity team's lower oil price projections reducing our 2026Q4 Brent forecast from \$90/bbl

to \$80/bbl, we expect the drag from elevated energy prices to gradually dissipate in the coming quarters.

2. **Slowdown in fiscal spending:** After January-February activity data beat expectations, the pace of fiscal spending slowed significantly in March and April. Combining the general public budget and government managed fund budget, total government expenditure growth dropped from +6.1% yoy in January-February to -7.3% yoy in April.¹ Other quasi-fiscal channels also showed tightening in recent months, with the PBOC's Pledged Supplementary Lending (PSL) balance contracting and policy bank bond net issuance falling. Now that April and May activity data have disappointed, we expect the pace of government bond issuance and fiscal spending to accelerate in the coming months (Exhibit 3).
3. **Adverse weather conditions and residual seasonality:** The NBS noted that heatwaves and heavy rainfall impeded both consumption and investment in May. Adverse weather conditions continued into June, likely weighing on outdoor activities. That said, June industrial production may benefit from positive quarter-end residual seasonality, and Q3 sequential growth should rebound if weather conditions normalize.²

Taken together, we expect Q3 sequential growth to pick up to 5.0% qoq annualized (vs. 4.5% previously) from 3.5% in Q2 (Exhibit 4). Our full-year real GDP forecast remains unchanged at 4.7% for 2026.

Exhibit 3: Local government special bond issuance slowed after Q1



Source: Wind, Goldman Sachs Global Investment Research

¹ Spending out of the general public budget and government managed fund budget totaled RMB40tn in 2025 (29% of GDP) and accounts for almost all of China's official government expenditure.

² According to the National Climate Center, the El Niño conditions may persist through the end of 2026, raising the odds of severe regional climate anomalies in Q3.

Exhibit 4: We nudge down Q2 sequential real GDP growth forecast while raising Q3 forecast

China real GDP growth forecast					
		New YoY%	Previous YoY%	New QoQ%, SAAR	Previous QoQ%, SAAR
2023		5.4	5.4		
2024		5.0	5.0		
2025		5.0	5.0		
2026		4.7	4.7		
2027		4.7	4.7		
2025	Q1	5.4	5.4	4.5	4.5
	Q2	5.2	5.2	4.5	4.5
	Q3	4.8	4.8	4.5	4.5
	Q4	4.5	4.5	4.9	4.9
2026	Q1	5.0	5.0	5.3	5.3
	Q2	4.5	4.7	3.5	4.0
	Q3	4.7	4.7	5.0	4.5
	Q4	4.6	4.6	4.5	4.5

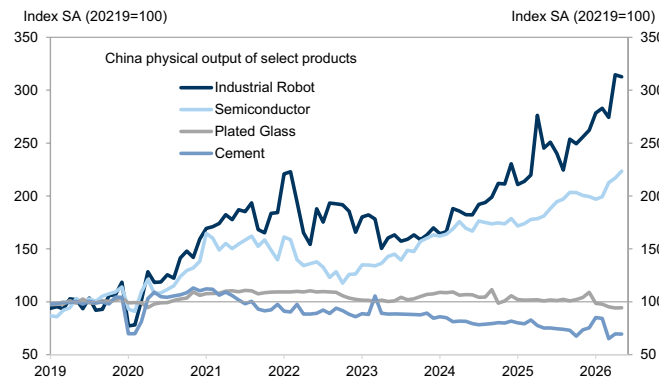
Source: NBS, Goldman Sachs Global Investment Research

Tech drives both the economy and the market

The Chinese government is restructuring the economy by shrinking the property sector and bolstering the technology sector. This transformation is clearly reflected in physical output data. Compared with 2019 levels, production of industrial robots has tripled, and semiconductor production has more than doubled, in sharp contrast to the decline in construction materials such as glass and cement ([Exhibit 5](#)). The divergence is also visible in equity market performance. The Shanghai Composite Index has been largely flat year-to-date, but beneath the stable aggregate index, the information technology sector index has rallied by over 50%, while the consumer discretionary sector index has slid by more than 25% ([Exhibit 6](#)).

Although policymakers have shown some concerns over the latest weakness in consumption growth, policy signals still point to unwavering support for tech and AI as China's structural transformation continues. During its press conference after the May data release, for example, the NBS touted China's technological advances in optical quantum computing and turbofan engine testing. According to NBS officials, high-tech manufacturing value-added grew 15.1% yoy, far outpacing overall IP growth of 4.5% yoy in May. In contrast to the 4.1% yoy decline in overall FAI in January-May, investment in the information transmission industry (which includes AI computing), R&D, and high-tech sectors increased 30.4% yoy, 9.3% yoy, and 4.5% yoy, respectively.

Exhibit 5: Physical output of industrial robots and semiconductors climbed while that of glass and cement fell



Source: CEIC, Goldman Sachs Global Investment Research

Exhibit 6: In the onshore equity market, tech index rallied while consumer index slid



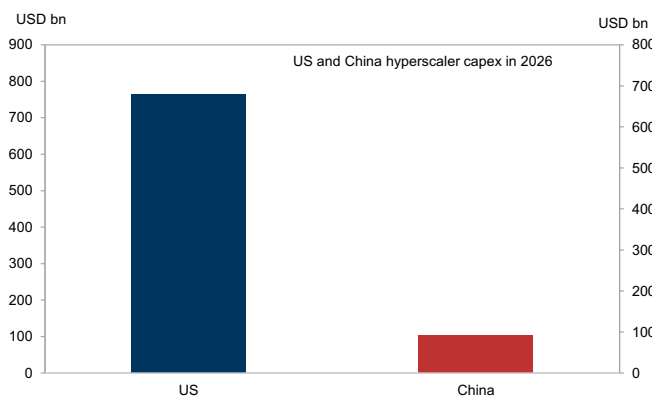
Source: Bloomberg, Goldman Sachs Global Investment Research

Racing for AI in a different way

On the surface, China's AI capex appears to fall far short of that in the US. For example, while US hyperscalers Amazon, Meta, Google, Microsoft, and Oracle are estimated to increase capex to over \$750bn in 2026, Chinese hyperscalers Alibaba, Tencent, ByteDance, and Baidu are expected to reach only \$100bn in total capex this year ([Exhibit 7](#)). However, this does not mean China's computing power is only a fraction of that in the US. As of mid-2025, installed data center capacity in China was already 60% of that in the US ([Exhibit 8](#)).

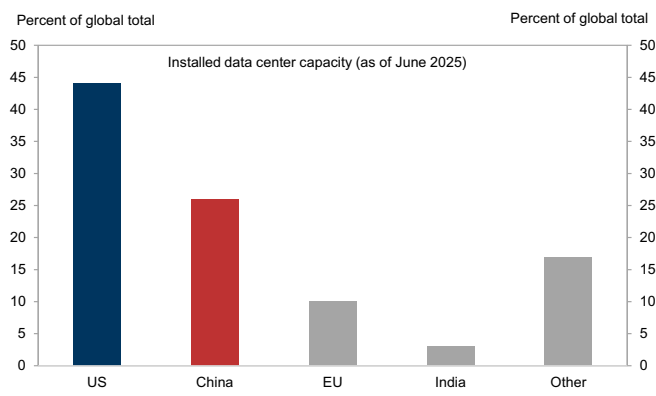
Two factors contributed to China's low hyperscaler capex relative to its data center capacity. First, the cost of building data centers is much lower in China than in the US. Second, a significant share of China's AI-related capex is conducted by the government. In the 15th Five-Year Plan (2026-30), for instance, policymakers plan to invest RMB2tn in computing power networks, including data centers. China's push for AI development and adoption is real, even though the way it is financed and applied differs significantly from the US.

Exhibit 7: Hyperscaler capex is much lower in China than in the US



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 8: China's installed data center capacity was 60% of that in the US as of June 2025



Source: IEA, Goldman Sachs Global Investment Research

As China continues to push aggressively on tech and AI advancement, investors are

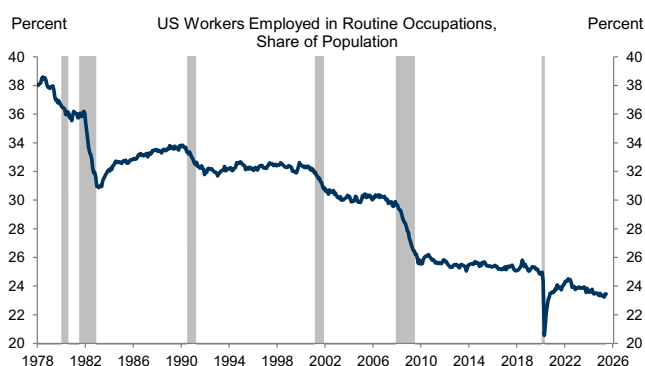
increasingly asking what this means for China's employment and consumption outlook. Regarding the pace at which AI adoption may replace labor, there are arguments on both sides. The case for slower AI adoption and less job displacement in China than in the US rests on relatively cheap labor and a government with more levers to discourage AI-related job losses in order to ensure social stability.

On the other hand, historical experience suggests that tech adoption to reduce labor costs accelerates during economic downturns, when profits erode and pressure on employers mounts. For example, our Global Economics team's research shows that, historically, nearly all job losses from automation of routine occupations in the US have been concentrated in economic downturns (Exhibit 9). With fierce competition persisting and many companies facing low margins, Chinese employers may be more willing than their US counterparts to replace labor with AI to stay competitive.

Herein lies the risk that too rapid a spread of tech and AI applications puts additional pressure on China's already weak labor market. After five years of decline, house prices are getting close to finding a bottom in the next year or two, based on the experience of historical international housing busts. However, if AI adoption were to lead to significant job losses, stagnant or even falling income would imply declining rents and further downward adjustment in house prices. In other words, with the domestic economy still weak and confidence fragile, another negative shock such as AI-related job displacement could amplify existing macro headwinds and delay the recovery in the property market and household consumption.

In June, the State Council released a document entitled "Implementation of the 15th Five-Year Plan for the Employment-First Strategy", which tries to strike a balance between accelerating China's AI development and ensuring labor market stability during 2026-30. However, it is unclear what concrete and effective measures could be introduced to create more jobs when the youth unemployment rate is already high and AI adoption threatens an increasing number of entry-level white-collar positions. Anecdotal, online job postings data suggest that growth in entry-level positions is underperforming the overall job market.

Exhibit 9: US experience shows that labor replacement tends to accelerate during economic downturns



Routine occupations include sales (e.g., clerks/telemarketers), office/administrative support (e.g., secretaries/typists), production (e.g., factory machine operators), construction/extraction (e.g., miners), and installation/maintenance/repair (e.g., mechanics).

Source: Jaimovich Siu 2018, Goldman Sachs Global Investment Research

Exhibit 10: Rising unemployment often precedes falling rents in China



Source: Haver Analytics, Goldman Sachs Global Investment Research

Policymakers focusing on the long-term

Recent policy news has focused on long-term strategies. The State Council Decree 834, published in April, addressed supply chain security. The State Council Decree 837, made public at the beginning of June, relates to monitoring and regulating tech transfers and cross-border investment amid US-China competition and Chinese companies' global expansion. The Lujiazui Forum in mid-June announced steps to build a price-based monetary policy framework and accelerate RMB internationalization.

In contrast, near-term cyclical policies have attracted less attention. But the near term is the path to the long term. We think three aspects of cyclical economic management are important in this regard.

1. As discussed earlier, if rapid AI adoption were to lead to significant job replacement, it would disrupt China's pursuit of a self-sustaining, robust, tech-driven growth model. Therefore, concrete and effective policies to boost employment and income are crucial to creating a domestic environment conducive to continued tech development.
2. With Chinese exports poised for strong growth in the next few years, the current economic planning system, which targets only headline real GDP growth, effectively caps domestic demand growth. This runs contrary to the government's long-term goal of expanding domestic demand. Instead, domestically oriented targets such as employment and consumption growth would be more suitable for China's stated long-term strategic objective of expanding domestic demand.
3. Confidence-building takes time and persistence. Over the past few years, we have observed a repeated pattern in which a solid quarter of real GDP growth is followed by a slowdown in fiscal spending, and policy easing accelerates only after weak data put the full-year target at risk. Quarter-to-quarter shifts in the policy stance do not help boost sentiment and bolster confidence, which are important to households' propensity to consume and private businesses' propensity to invest, even if such shifts help the government land official GDP statistics exactly on the growth target laid out at the beginning of the year.

The experience of the past two years, including the US-China trade war of 2025 and the energy supply shock of 2026, has demonstrated the advantages of China's long-term planning and strategic thinking. However, if long-term planning becomes policymakers' sole focus, it could hinder progress toward achieving strategic goals. Fostering consistent and strong economic growth through cyclical tools should be an indispensable part of macro management. Robust growth, a strong labor market, and optimistic sentiment provide the cushion needed for the government to push through difficult structural changes and enable a virtuous cycle between tech-driven productivity gains and strong domestic demand growth.

Hui Shan

The China Economics Team

Andrew Tilton

+852-2978-1802
andrew.tilton@gs.com
Goldman Sachs (Asia) L.L.C.

Hui Shan

+852-2978-6634
hui.shan@gs.com
Goldman Sachs (Asia) L.L.C.

Lisheng Wang

+852-3966-4004
lisheng.wang@gs.com
Goldman Sachs (Asia) L.L.C.

Xinquan Chen

+852-2978-2418
xinquan.chen@gs.com
Goldman Sachs (Asia) L.L.C.

Yuting Yang

+852-2978-7283
yuting.yyang@gs.com
Goldman Sachs (Asia) L.L.C.

Chelsea Song

+852-2978-0106
chelsea.song@gs.com
Goldman Sachs (Asia) L.L.C.

Disclosure Appendix

Reg AC

I, Hui Shan, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Hui Shan Goldman Sachs (Asia) L.L.C..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered

companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.